

1. (a) (2 points) Let $p > 0$. Evaluate the integral

$$\int_a^1 \frac{1}{x^p} dx,$$

where $0 < a < 1$.

- (b) (3 points) Let $p > 0$. For what values of p will the improper integral

$$\int_0^1 \frac{1}{x^p} dx$$

be convergent? For the values of p where the improper integral is convergent, evaluate the improper integral. (Recall that $\int_0^1 \frac{1}{x^p} dx = \lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x^p} dx$.)

$$\begin{aligned} \int_a^1 \frac{1}{x^p} &= \int_a^1 x^{-p} \\ &= \left[\frac{x^{-p+1}}{-p+1} \right]_a^1 \end{aligned}$$

$$= ?$$

$$b) \left[\frac{x^{-p+1}}{-p+1} \right]_a^1$$

~~p must be equal to 1 to be improper integral~~

~~p must be more than 1 to be convergent~~

$$c) \int_0^1 \frac{1}{x^p} dx \quad \text{since } x \neq 0$$

$$\lim_{a \rightarrow 0^+} \int_a^1 \frac{1}{x^p} dx$$

$$\lim_{a \rightarrow 0^+} \left[\frac{x^{-p+1}}{-p+1} \right]_a^1$$

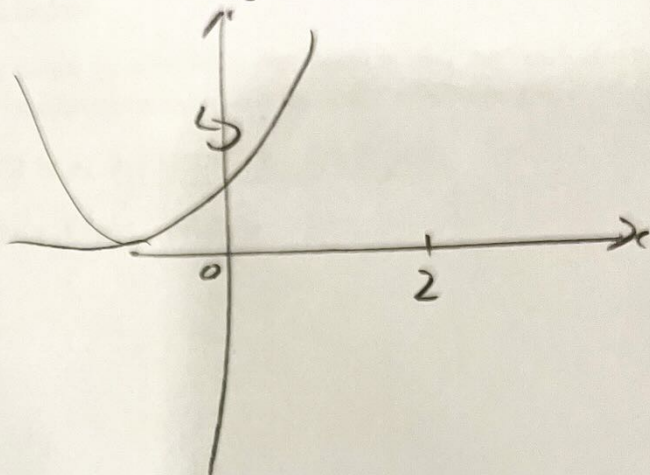
$$= \frac{1^{-p+1}}{-p+1} \quad p > 1$$

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2. (5 points) Find the volume of the solid obtained by rotating the region bounded by $y = 1 - (x-1)^2$ and $y = 0$ about the y -axis. Give exact answer.

Shell method bounded $y = 1 - (x-1)^2$, $y = 0$ about ~~x~~ y -axis

$2\pi \int_0^2 x(1-(x-1)^2) dx$ height = $1-(x-1)^2$
radius = x



$$\text{vol} = 2\pi \int_0^2 x(1-(x-1)^2) dx$$

$$= 2\pi \int_0^2 x(2x - x^2) dx$$

$$= 2\pi \int_0^2 (2x^2 - x^3) dx$$

$$= 2\pi \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2$$

$$= 2\pi \left[\frac{16}{3} - \frac{32}{4} \right] - [0]$$

$$= 2\pi \left(\frac{32}{12} \right)$$

$$= \frac{32}{3} \pi$$

$$= \frac{16}{3} \pi$$

$$\begin{aligned} (x-1)(x-1) \\ x^2 - x - x + 1 \\ = x^2 - 2x + 1 \end{aligned}$$

$$\begin{aligned} y &= 1 - (x^2 - 2x + 1) \\ &= 1 - x^2 + 2x - 1 \\ &= 2x - x^2 \end{aligned}$$

$$= x(2-x)$$

$$x=0, x=2$$

NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

TCX2101 Calculus and Linear Algebra for Computing

Class Test 2

1. Write your answers in the space provided below.
2. To submit, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file (make sure they are in order of the page).
3. Name the PDF file by StudentNo_CT2 (e.g. A1234567Z_CT2.pdf).
4. Submit the PDF file to Canvas assignments, Class Test 2.

Student No.:

A	0	3	3	6	0	5	0	R
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